INCH-POUND
MIL-C-24746(SH)
31 December 1990

MILITARY SPECIFICATION

COOLERS, UNIT FORCED AIR FOR SHIP'S REFRIGERATED STORES

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes forced air unit coolers for use in food storage chill and freeze rooms on board Naval ships.

1.2 Classification. Unit coolers shall be of the following sizes and capacities (see 3.4 and 6.2):

Standard size	Standard capacity (tons)	British thermal units per hour (Btu/hr)
I	1/8	1,500
II	1/4	3,000
III	3/8	4,500
IV	1/2	6,000
V	3/4	9,000
VI	1	12,000
VII	1-1/2	18,000

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.
QQ-B-654 - Brazing Alloys, Silver.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/6 - Plates, Identification, Equipment.
MIL-P-15280 - Plastic Material, Unicellular (Sheets and Tubes).
MIL-R-16743 - Refrigerating Plants and Systems, Mechanical; and Refrigeration System Components - Dichlorodifluoromethane Type-12 - Naval Shipboard.
MIL-M-17059 - Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use).
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-E-19933 - Electrodes and Rods - Welding, Bare, Chromium and Chromium-Nickel Steels.
MIL-E-22200/2 - Electrodes, Welding, Covered (Austenitic Chromium-Nickel Steel).
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-V-23450 - Valves, Expansion, Thermostatic, Refrigerant 12 and Refrigerant 22.
MIL-E-24142 - Enclosures For Electrical Fittings and Fixtures, General Specification For.
MIL-P-24423 - Propulsion and Auxiliary Control Consoles and Associated Control and Instrumentation Equipment, Naval Shipboard Use, Basic Design Requirements.
DOD-G-24508 - Grease, High Performance, Multi-Purpose. (Metric)
MIL-R-46089 - Rubber, Sponge, Silicone, Closed Cell.
MIL-C-81302 - Cleaning Compound, Solvent, Trichlorotrifluoroethane.
MIL-T-81955 - Tin Plating; Immersion for Copper and Copper Alloys.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.

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MILITARY (Continued)

MIL-STD-278 - Welding and Casting Standard.
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-785 - Reliability Program for Systems, and Equipment Development and Production.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1399, Section 301 - Interface Standard for Shipboard Systems Ship Motion and Attitude. (Metric)
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
803-6397311 - Standard Dimensions for Coolers Forced Air Unit Type for Ship's Refrigerated Stores.

(Application for copies should be addressed to: Commander, Portsmouth Naval Shipyard, Naval Engineering Drawing Support Activity, Code 202.2, Portsmouth, NH, 03804-5000.)

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Application for copies should be addressed to the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

Bulletin 210 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to Air Movement and Control, Inc., 30 West University Drive, Arlington Heights, IL 60004.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 240 - Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip For Pressure Vessels. (DoD adopted)

A 269 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service. (DoD adopted)

A 276 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes.

A 366 - Standard Specification for Steel, Sheet, Carbon, Cold-Rolled, Commercial Quality. (DoD adopted)

A 789 - Standard Specification for Seamless and Welded Ferritic/ Austenitic Stainless Steel Tubing for General Service.

B 68 - Standard Specification for Seamless Copper Tube, Bright Annealed.

B 88 - Standard Specification for Seamless Copper Water Tube. (DoD adopted)

B 108 - Standard Specification for Aluminum Alloy Permanent Mold Castings. (DoD adopted)

B 111 - Standard Specification for Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock. (DoD adopted)

B 209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate.

B 395 - Standard Specification for U-Bend Seamless Copper and Copper Alloy Heat Exchanger and Condenser Tubes. (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

F 593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs. (DoD adopted)

F 594 - Standard Specification for Stainless Steel Nuts. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

Standard 25 - Methods of Testing Forced Convection and Natural Convection Air Coolers for Refrigeration.

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air-Conditioning Engineers, 1791 Tullie Circle NE, Atlanta, GA 30329.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ANSI/ASME B31.5 - Refrigeration Piping.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

UNDERWRITERS LABORATORIES INC. (UL)
412 - Refrigeration Unit Coolers.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Reliability. The equipment shall have an operating life of at least 40,000 hours. The forced air unit coolers supplied by this specification shall have a mean-time-between-failures (MTBF) of 8,000 hours, mean-time-to-repair (MTTR) of 8 hours, mean-time-between-corrective-maintenance (MTBCM) of 8,000 hours, and an operational availability (A_o) of 0.95 (see 4.6.14.6).

3.3 Materials. Materials for use in the construction of the unit coolers shall be as specified in table I, or as otherwise specified (see 6.2).

TABLE I. Materials and processes.

Application	Materials	Specification
Unit cooler		
Casing, frame mounting brackets, and fasteners (nuts and bolts)	302 or 304 stainless steel	ASTM A 240, A 269, A 276 F 593, F 594 as applicable
Louver	302 or 304 stainless steel	ASTM A 240, A 269, A 276 F 593, F 594 as applicable
Drain pan	302 or 304 stainless steel	ASTM A 240, A 269, A 276 F 593, F 594 as applicable

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TABLE I. Materials and processes - Continued.

Application	Materials	Specification
Identification plate	302 or 304 stainless steel	ASTM A 240, A 269, A 276 F 593, F 594 as applicable
Welding electrodes and rods	Low carbon steel	MIL-E-19933, MIL-E-22200/2
Insulation		MIL-P-15280
Insulation adhesive		MIL-P-15280
Cooling coil		
Tubes	Seamless copper tube	ASTM B 68
Suction header	Seamless copper tube	ASTM B 68
Fins	3000 or 5000 series aluminum alloy	ASTM B 209
Tube sheets	302 or 304 stainless steel	ASTM A 240
Ferrules	Copper or brass	ASTM B 111
Brazing	Silver brazing alloy or phosphorous copper	QQ-B-654
Refrigerant	Type R-12 or R-22	BB-F-1421
Fan assembly		
Fan blade and hub	Bronze, aluminum alloy or corrosion resistant steel	ASTM B 108
Fan guard	302 or 304 stainless steel	ASTM A 240, A 269, A 276 F 593, F 594
Electrical enclosures		
Controls enclosure remote to unit cooler	ASTM A 366 and ASTM A 789	MIL-E-24142
Junction box attached to unit cooler	304 stainless steel	MIL-E-24142
Gaskets	Silicon rubber	MIL-R-46089

3.3.1 Materials not specified. When a definite material is not specified, a material shall be used which will enable the unit cooler to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product. Materials shall be free from defects and suitable for the purpose intended.

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3.3.2 Dissimilar metals. Direct contact of electrolytically dissimilar metals, as defined and classified in accordance with MIL-STD-889, shall be prohibited.

3.3.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Size and capacity. Sizes and capacities of unit coolers shall be as described in 1.2 or as specified (see 6.2). The capacities shall be based on the following design conditions.

3.4.1 Air-on temperature. Unit coolers shall meet the rated capacity with an air-on (inlet air) temperature of minus 10 degrees Fahrenheit (°F) and an evaporator (refrigerant) temperature of minus 20°F.

3.4.2 Liquid refrigerant temperature. For coil and expansion valve sizing, the liquid refrigerant temperature at the inlet of the expansion valve shall be taken as 103°F with a corresponding pressure of 123 pounds per square inch (lb/in^2) gauge.

3.4.3 Air velocity. Air velocity across the face of the coil shall be not greater than 600 feet per minute (ft/min).

3.4.4 Convertibility. Unit coolers shall be convertible from frozen storage application to chilled storage. Design storage temperatures for frozen storage and chilled storage shall be minus 10°F and plus 33°F respectively.

3.5 Construction. Forced air unit coolers shall consist of a casing, evaporative cooling coil, fan and motor assemblies, and louvers located on the coil side of the unit cooler which close during the defrost cycle, built as a single unit, ready for installation and connection to the refrigerant supply and return lines, and condensate drain line (see 6.3). Each unit cooler shall have electric defrost which shall consist of electric heating elements in direct contact with the finned surface, drain pan, and drain line. The cooling and defrost controls shall be combined in a single enclosure, separate from the unit cooler, and ready for connection to the power supply and the unit cooler.

3.6 Shock. Unit coolers and controls shall withstand the high impact shock requirements in accordance with MIL-S-901 for grade A, class I equipment.

3.7 Vibration. Unit coolers and controls shall withstand the type I (environmental) vibration in accordance with MIL-STD-167-1 encountered on board ship.

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3.8 Combination of list, trim, roll, and pitch. Unit coolers shall perform under the list, trim, roll, and pitch conditions as specified in DOD-STD-1399, section 301. The unit cooler shall show no signs of spillage or dripping of condensate when tilted 30 degrees in any direction.

3.9 Casing. The unit cooler casing shall consist of a fan/coil enclosure, mounting brackets, drain pan, and directional louvers, and shall be fabricated to support all the components that make up the unit cooler assembly. Fasteners, sheet metal, framing and mounting brackets shall be in accordance with table I. Physical dimensions of the unit cooler enclosure and mounting brackets, and the maximum weights shall be in accordance with Drawing 803-6397311.

3.9.1 Drain pan. The drain pan shall be removable for routine cleaning. It shall be insulated to prevent condensation and provided with heaters to ensure drainage during the defrost cycle. The heaters shall be in accordance with 3.15.10. The insulation shall have a resistance (R-value) of not less than $3.7 hr \cdot ft^2 \cdot °F/Btu$, and shall be bonded in place with insulation and adhesive not greater than 1 inch total thickness. The drain pan shall continue to drain when tilted in any direction to the condition of list specified in 3.8. Materials shall be in accordance with table I. The drain line connections at the drain pan shall be as shown on Drawing 803-6397311.

3.9.2 Directional louvers. Louvers shall be of vertical configuration and shall be adjustable. The louvers shall be lockable once adjusted. Material shall be in accordance with table I.

3.9.3 Defrosting louvers. These louvers shall be of a horizontal configuration designed to open by the air flow and close when the fans are stopped. They shall be located on the inlet side of draw through coolers and on the outlet side of blow through coolers.

3.10 Cooling coil. The cooling coil shall be a direct expansion, extended surface type for use with refrigerant R-12 or R-22 as specified (see 6.2). It shall be constructed of copper tubing expanded into aluminum fins. An expansion valve, distributor, and suction header shall also be provided. The suction and liquid lines shall be as shown on Drawing 803-6397311 and shall be secured to the unit cooler frame. Unless otherwise specified (see 6.2), the coil shall be right-hand as shown on Drawing 803-6397311.

3.10.1 Tubes and suction header. Tube wall thickness and suction header wall thickness shall be type L of ASTM B 88 or as specified in ANSI/ASME B31.5, whichever is greater. U-bends shall have an internal diameter at least equal to the diameter of the tube as specified in ASTM B 395, and the minimum wall thickness shall be maintained throughout the bend. Tube and header material shall be in accordance with table I. The suction header shall not entrap oil. Coils shall not leak (see 4.6.6).

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3.10.2 Fins. Fins shall be of the continuous plate type, not less than 0.010 inch thick in accordance with table I. Fins shall have rolled collars with tubes expanded into the collars to form a permanent bond. The number of fins per linear inch shall be not less than three nor greater than five, and they shall be uniformly spaced.

3.10.3 Tube sheets and intermediate tube sheets. Tube sheets and intermediate tube sheets shall support the cooling coil inside the unit cooler casing. Spacing between tube supports shall be not greater than 24 inches. Tube sheet material shall be in accordance with table I.

3.10.4 Ferrules. Ferrules shall be provided where tubes pass through tube sheets and intermediate tube sheets in order to prevent wear on the tubes caused by expansion and contraction of the tubes and shipboard vibration. Tubes shall be expanded into the ferrules to ensure that the wear area is between the ferrules and tube sheets.

3.10.5 Expansion valve. The expansion valve shall be in accordance with MIL-V-23450 and shall be of the modulating type actuated by a direct operating thermal type element. The capillary tube shall be 5 feet long. The expansion valve shall be externally equalized. Disassembly of the expansion valve for repair or replacement of parts shall be possible without disconnecting other than the external equalizer. The expansion valve shall be located inside the unit cooler, and shall be not greater than 12 inches from the distributor inlet.

3.10.6 Distributor. A distributor shall be supplied to ensure proper distribution of refrigerant to each circuit of the coil. The distributor shall be manufactured as a part separate from the expansion valve. Outlet connections to each circuit shall be in accordance with the following requirements:

(a) Three outlets 120 degrees apart for coils having three circuits or less. The unused outlets shall be plugged.
(b) Six outlets 60 degrees apart for coils having five or six circuits. The unused outlets shall be plugged.
(c) Twelve outlets 30 degrees apart for coils having twelve circuits or multiples thereof. The unused outlets shall be plugged.
(d) Wall thickness in accordance with 3.10.1.

3.10.7 Brazing. Piping connections inside the unit cooler shall be brazed in accordance with NAVSEA 0900-LP-001-7000. Brazed joints shall be in accordance with MIL-STD-278 and table I. Brazing procedures and the qualification of the brazer shall be in accordance with MIL-STD-248.

3.11 Serviceability. The unit cooler shall be accessible from both sides and the front for servicing the fans, motors, heating elements, and wiring. The maintenance envelope shall be as defined on Drawing 803-6397311. The heating elements shall be readily accessible from both sides of the unit cooler and shall be replaceable within the maximum maintenance envelope. Maintenance of the unit cooler shall be possible without the use of special tools (see 3.17).

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3.12 Fans. The fans for the unit cooler shall be propeller type axial fans, complete with motor, mounting plate, and fan guard when required. Fans shall be direct-driven with the fan wheel overhung on the motor shaft. Multiple fans shall be used and shall be of the same manufacturer, model, cubic feet per minute (ft^3/min), and power ratings. The fans shall provide a velocity across the face of the coil as specified in 3.4.3. Fans shall run at all times while the unit cooler is in service, except during the defrost cycle. Fan assemblies shall be balanced to the type II (internally excited) vibration requirements as specified in MIL-STD-167-1 (see 4.6.12).

3.12.1 Fan motor. The fan motor shall be a totally enclosed air over motor that meets the requirements for minus 20°F service conditions. It shall be supplied with low temperature grease and a corrosion-resistant steel shaft. The fan motor shall be in accordance with MIL-M-17059 for service A application and shall have a sealed insulation system. The motor shall withstand a plus or minus 10 percent variance in the rated voltage without measurable damage. The motor shall also be provided with internal thermal overload protection with automatic reset. The fan motor shall be mounted on resilient mounts so that it is dynamically isolated from the unit cooler. Lubrication of the motor shall meet the requirements of DOD-G-24508.

3.12.2 Fan blades and hub. Material for the fan blades and hub shall be in accordance with table I. Fan blades shall be attached to the hub by riveting or welding, or the fan blades and hub may be formed, cast, or molded as one piece from the same material stock. The fan blade and hub shall be keyed to the motor shaft and held in place by double, self-locking set screws 90 degrees apart. Fan blade and hub assemblies shall be single plane balanced to within plus or minus 0.10 ounce per inch (oz/in) (see 4.6.9).

3.12.3 Fan location. Fans shall be located inside the unit cooler so that it serves as a draw-through device. The fan wheel may project from the unit cooler casing only if the blades are totally encased in the fan guard. See Drawing 803-6397311 for the permissible projection of the fan wheel and guard.

3.12.4 Fan guard. A fan guard shall be provided to prevent injury by the fan blades. The fan guard shall be in accordance with the following requirements:

(a) Materials in accordance with table I.
(b) Construction to permit ready removal.
(c) Compliance with UL 412 for protection of personnel from injury by moving parts.
(d) No appreciable restriction of air flow.
(e) No vibration when the fan is in operation.
(f) Construction that prevents distortion under normal operating conditions.

The fan guard may be incorporated into the construction of the directional louvers (see 3.9.2), providing the above requirements are met.

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3.13 Airborne noise. Unit coolers and controls furnished under this specification shall meet grade D noise requirements when tested in accordance with MIL-STD-740-1 (see 4.6.13).

3.14 Structureborne noise. When specified (see 6.2), unit coolers and controls shall meet the structureborne noise requirements (for example, type III) in accordance with MIL-STD-740-2.

3.15 Electrical. Electrical equipment shall be in accordance with the requirements of MIL-P-24423.

3.15.1 Electrical service. The unit cooler shall receive the ship's 440 volt, 3 phase, 60 hertz (Hz) electrical service at the controls enclosure (see 3.15.3.1). Power reduction for lower voltage, single phase fan motor, heaters, and controls shall occur inside the controls enclosure. The service connection at the unit cooler shall be located as shown on Drawing 803-6397311. Heater power shall be supplied by the 440 volt, 3 phase, 60 Hz supply, connected as a balanced load.

3.15.2 Electrical connections. Connections or terminations shall be clearly and indelibly numbered or otherwise marked on terminal strips. Electrical connections or terminations shall be made on screw type terminal strips.

3.15.3 Electrical enclosures. Electrical enclosures shall be totally enclosed and splashproof. Enclosures shall be constructed as specified in MIL-E-24142 for nonsubmersible enclosures. Wire penetrations shall also be splashproof.

3.15.3.1 Controls enclosure. Cooling controls and defrost controls shall be housed in the same enclosure and shall be supplied separate from the unit cooler. They shall be provided with a location outside the refrigerated space. A numbered terminal block shall be provided inside the enclosure with a corresponding terminal block located at the unit cooler. The enclosure shall contain a fused power disconnect suitable for the application intended (see 3.15.1).

3.15.3.2 Electrical enclosure at unit cooler. Materials for the electrical enclosure (junction box) at the unit cooler shall be the same as those specified (see table I) for the unit cooler enclosure.

3.15.4 Indicating lights. Indicating lights shall be provided to indicate the following functions:

POWER ON	white
DEFROST INHIBITED	blue
DEFROST	green
HEATER FAILURE	red
DEFROST TROUBLE	amber

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The heater failure and defrost trouble indicators shall require manual reset. Indicating lights shall be located on the front of the controls enclosure, with the words "POWER ON", "DEFROST INHIBITED", "DEFROST", "HEATER FAILURE" and "DEFROST TROUBLE" printed directly below the respective light. A terminal strip shall be supplied inside the controls enclosure for the connection of remote indicating lights. A circuit with a momentary action type switch shall be provided to test the indicating lights.

3.15.5 Electric shock hazard. The unit cooler shall not be a source of electric shock hazard (see 4.6.11).

3.15.6 MANUAL-OFF-AUTO mode switch. A three-position rotary control switch shall be provided for each unit cooler. It shall be mounted on the front of the controls enclosure. Switch positions shall be MANUAL-OFF-AUTO and shall be so marked.

3.15.6.1 MANUAL position. The MANUAL position shall permit cooling by the unit cooler. With the mode switch in this position, the only means of initiating the defrost cycle shall be by a push button switch (see 3.15.8.1). The push button switch shall be labeled "MANUAL DEFROST" and shall be operative only when the three-position switch is in the MANUAL position.

3.15.6.2 OFF position. The OFF position shall deenergize the unit cooler, refrigerant solenoid valve, defrost controls, heater elements, and associated circuitry.

3.15.6.3 AUTO position. The AUTO position shall permit cooling and shall permit the defrost cycle of the unit cooler to be automatically initiated by a control timer (see 3.15.8.1). The control timer shall totalize the refrigerant valve's ON time and shall be an interval timer that is adjustable from 0 to 90 minutes. A manual override switch shall be provided on the timer to initiate the output control signal.

3.15.7 Cooling control. The cooling control for each unit cooler shall be provided by a temperature controller that cycles the refrigerant solenoid valve.

3.15.7.1 Temperature controller. The temperature controller shall be adjustable from minus 15 to plus 38°F. The sensing element shall be located in the inlet air stream of the unit cooler. The adjustment for the temperature controller shall be temperature indicated and set by screwdriver, and shall be located in the controls enclosure. Accuracy of the setting shall be plus or minus 2°F of the indicated temperature.

3.15.7.2 Refrigerant solenoid valve interface. A 110 volt single phase, 60 Hz power supply for the refrigerant solenoid valve shall be provided at the controls enclosure. Terminals (see 3.15.2) shall be provided and marked "Refrigerant Solenoid Valve" for the interface connections. The power supply shall deliver to the solenoid an in-rush current of 0.33 ampere and a holding current of 0.17 ampere. The refrigerant solenoid valve shall meet the requirements of MIL-R-16743 and shall be supplied by the contractor unless otherwise specified (see 6.2).

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3.15.7.3 Elapsed timer. A 100,000-hour elapsed time meter shall be provided for monitoring the unit cooler's operating hours. The elapsed time meter shall measure the ON time of the refrigerant solenoid valve in hours.

3.15.8 Electric defrost control. The electric defrost control shall consist of a system to dictate the sequence of events of defrost as specified in 3.15.8.1 through 3.15.8.3.

3.15.8.1 Defrost initiation. In the AUTO mode, defrost shall be initiated by a control timer that totalizes the ON time of the refrigerant solenoid valve; that is, the timer shall clock the cooling time of the unit cooler (see 3.15.6.3). In the MANUAL mode, defrost shall be initiated by a push button switch (see 3.15.6.1). Upon initiation of defrost, the unit cooler shall be inhibited from defrosting unless there are no other unit coolers in defrost. If the unit cooler is inhibited from defrosting by a unit already in defrost, it shall continue cooling (as stores require) until a permissive signal is received. A master sequencer shall be provided for the permissive/inhibited function as follows:

(a) The next unit cooler in sequence waiting to defrost shall be permitted to begin its defrost cycle while all other unit coolers in the system are inhibited from beginning their defrost cycle.
(b) The master sequencer shall provide for the interfacing of four unit coolers by delegating the defrost permissive inhibit functions.
(c) A master sequencer shall be provided in each control enclosure. One of the control units shall be designated the master sequencer on installation. Unused master sequencers shall be considered spares.
(d) The master sequencers shall not default the control timer to inhibit defrost, except that the control timer may be defaulted once it has completely timed out.
(e) When defrost is inhibited, the condition shall be made known by a blue indicating light.

3.15.8.2 Defrost cycle. When permission is given for the unit cooler to defrost, the defrost controls shall cycle through the following sequence of events:

(a) All other unit coolers shall be inhibited from defrosting until the defrost cycle for the one permitted to defrost is completed.
(b) A green indicating light shall be energized to indicate the unit cooler in defrost.
(c) The unit cooler's fans shall be deenergized and locked out until the defrost cycle is completed. The fans shall be operating at all times except during defrost.
(d) The refrigerant solenoid valve shall be closed to stop the refrigerant flow through (and subsequent cooling of) the unit cooler's coil.

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(e) The coil heating elements shall be energized to defrost the coil.
(f) The drain pan heating elements and the drain line heating cable shall be energized to enable drainage of the condensate from the unit cooler.

3.15.8.3 Defrost termination. When the temperature sensed at the coil surface has risen to a point that will ensure total defrost, the controls shall simultaneously:

(a) Deenergize the coil, drain pan, and drain line heating elements.
(b) Energize the unit cooler fans.

After 1 minute of fan operation, the following shall occur simultaneously:

(c) Open the refrigerant solenoid valve to permit the coil to return to its operating temperature.
(d) Reset the control timer to begin totalizing the refrigerant solenoid valve's ON time in order to determine the next defrost cycle. The timer shall be inoperative in the manual mode.
(e) Deenergize the green, defrost indicating light. The defrost cycle has now been completed.
(f) The master sequencer shall now seek out and permit the next unit cooler in sequence that is waiting to defrost to begin its defrost cycle.

3.15.9 Back-up defrost termination. A back-up defrost termination shall be provided by an interval timer that begins timing when the coil heating elements are energized. If the timer totalizes 20 minutes before the defrost cycle is completed, it shall automatically put the defrost cycle through steps 3.15.8.3(a), (b), (c), (d), and (e) simultaneously. If this timer terminates defrost, an amber (DEFROST TROUBLE) indicating light shall be energized until it is manually reset. The back-up defrost termination timer shall be continuously adjustable from 0 to 30 minutes by screwdriver adjustment.

3.15.10 Heater elements. Heater elements for the coil and drip pan shall be stainless steel, sheathed, continuous in length, and in accordance with MIL-H-22577. The serviceability of the heater elements shall be in accordance with 3.11. Heater elements in direct contact with the finned surface shall be sized to melt a buildup of 0.1 inch average thickness of ice between the fins within 8 minutes. Unless otherwise specified (see 6.2), the unit cooler shall be supplied with a quantity of heating cable to wrap 12 feet of 1-1/4 inch diameter drain pipe. The cable shall provide, when spirally wrapped, 125 $\pm$ 6 Btu/hr of heating per linear foot of pipe. Heaters shall be wired to a balanced load (see 3.15.1).

3.15.10.1 Heater failure detection. The heater firing circuit shall be monitored for determination of heater failure. The condition of heater failure shall be made known by an indicating light (see 3.15.4).

3.15.11 Wiring. The interface wiring between the controls enclosures and unit coolers, and the master sequencer and other control enclosures shall not be accomplished by the contractor.

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3.16 Identification plates. Each unit cooler and defrost controls enclosure shall be provided with an identification plate conforming to MIL-P-15024 and MIL-P-15024/6. The material for the identification plate shall be the same as that used for the casing, and the plate shall be located as shown on Drawing 803-6397311. In addition to the requirements of MIL-P-15024 and MIL-P-15024/6, the identification plate shall contain the following information: unit cooler capacity, fan motor horse power, nominal ft^3/min, fan supply voltage, and unit cooler total power requirements.

3.17 Special tools. Special tools shall not be required to maintain the unit cooler. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

3.18 Workmanship. Cooling coils and unit coolers shall be free from defects that affect their appearance and operation. Tube sheets shall show no cracks resulting from punching or forming, and fin collars shall show no cracks after tubes are expanded into collars. Fin edges shall be free of burrs and shall not be bent. Casing corners shall be square, flanges shall be parallel and smooth, the fan guard screen shall not be bent, and mounting bolts for direction louvers and the fan assembly of the unit cooler shall be in place and secured. The unit cooler shall show no signs of excessive vibration through the start up and operation of the fan and motor assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), inspections shall be performed in accordance with the test conditions specified in 4.3.1 and 4.3.2.

4.3.1 Notification of tests. The contracting activity shall be given 2 weeks notice of all tests in order to be able to schedule and witness tests (see 6.3).

4.3.2 Instruments. Instruments used in the performance of the tests shall be calibrated before and after the tests, and the results of both calibrations shall be determined.

4.4 First article inspection. When first article inspection is required, first article inspection shall consist of the examinations and tests in accordance with table II. The sizes of forced air unit cooler assemblies to be inspected shall be as specified (see 6.2 and 6.3).

TABLE II. First article and quality conformance inspection.

	Requirement	Test method	First article	Quality conformance
Group A				
Visual and dimensional	3.3, 3.4, 3.5, 3.9, 3.10, 3.11, 3.12, 3.15, 3.16, 3.18	4.6.1	X	X
Inspection for cleaning	5.2	4.7	X	X
Inspection for packaging	5.1	4.7	X	X
Group B				
Capacity	3.4	4.6.2	X	
Leakage	3.10.1	4.6.6	X	X
Fan	3.12	4.6.7	X	
Motor	3.12.1	4.6.8	X	
Fan balance	3.12.2	4.6.9	X	X
Airborne noise	3.13	4.6.13	X	
Structureborne noise	3.14	6.2	X	

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TABLE II. First article and quality conformance inspection - Continued.

	Requirement	Test method	First article	Quality conformance
Leakage current	3.15.5	4.6.11.1	X	
Insulation resistance, unit cooler	3.15.5	4.6.11.2	X	X
Insulation resistance, heating elements	3.15.5	4.6.11.3	X	
Dielectric strength	3.15.5	4.6.11.4	X	X
Operation		4.6.14		
Fan operation	3.12	4.6.14.1	X	X
Defrost	3.15.8	4.6.14.2	X	X
Sequence	3.15.8.1	4.6.14.3	X	
Heater element	3.15.10	4.6.14.4	X	
Drainage	3.9.1	4.6.14.5	X	
Group C				
Shock	3.6	4.6.3	X	
Vibration	3.7	4.6.4	X	
List, trim, roll	3.8	4.6.5	X	
Splash	3.15.3	4.6.10	X	
Group D				
Reliability acceptance	3.2	4.6.14.6	X	1/

1/ Unless otherwise specified (see 6.2).

4.5 Quality conformance inspection. Unit coolers offered for delivery shall be inspected as specified in table II.

4.5.1 Noncompliance. If a sample fails to pass group C or D inspection, the contractor shall notify the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted. In addition, action shall be taken with respect to units of product which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure, if a correction can be made. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the activity, has been taken. After the corrective action has been taken, group C or D inspections, or both, shall be repeated on additional sample units (all tests and examinations or the test which the original sample failed, at the option of the contracting activity). Groups A and B inspections may be

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reinstituted; however, final acceptance and shipment shall be withheld until the group C or D inspection, or both, have shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the contracting activity.

4.6 Examinations and tests.

4.6.1 Visual and dimensional examination. Each unit cooler shall be visually and dimensionally examined in accordance with table III to verify compliance with the requirements of this specification. No defects are permitted for the classification of possible defects listed in table III. Any unit cooler containing one or more defects shall not be offered for delivery.

TABLE III. Visual and dimensional examination.

Defects	Requirements
1. Material not as specified or defective.	3.3, 3.3.1, 3.3.2
2. Size of unit cooler not as specified.	3.4
3. Weight of unit cooler exceeds maximum allowed.	3.9
4. Unit cooler casing not as specified; limiting dimensions exceeded.	3.9
5. Mounting dimensions not as specified.	3.9
6. Bolts, nuts, and screws not tight, or missing (parts shall be properly fastened and secured).	3.9
7. Drain pan missing or not as specified.	3.9.1
8. Drain line connection not as shown on referenced drawing.	3.9.1
9. Insulation missing or not as specified.	3.9.1
10. Louver missing or inoperative.	3.9.2
11. Cooling coil not as specified.	3.10
12. Missing expansion valve or distributor.	3.10, 3.10.6, 3.10.7
13. Suction and liquid lines not as shown on referenced drawing or coil and not as specified.	3.10
14. Suction and liquid lines not secured to unit cooler frame.	3.10
15. Suction header not as specified.	3.10, 3.10.5
16. Suction header entraps oil.	3.10.1
17. Tube construction not as specified.	3.10.1
18. U-bends not maintaining minimum wall thickness or internal diameter.	3.10.1
19. Fin thickness and spacing not as specified.	3.10.2
20. Tube sheets and intermediate tube sheets exceed 24-inch minimum spacing.	3.10.4
21. Fin collars, tube sheets, and intermediate tube sheets not free of cracks or burrs.	3.10.4
22. Ferrules missing or not as specified.	3.10.5
23. Brazing material not as specified or brazing of poor quality.	3.10.8

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TABLE III. Visual and dimensional examination - Continued.

Defects	Requirements
24. Unit cooler not accessible from both sides and front.	3.11
25. Wrong type or missing fan motor.	3.12, 3.12.1
26. Fan blade missing or improper fan blade attachment to hub.	3.12, 3.12.2
27. Fan not located in exiting air stream.	3.12.3
28. Fan guard not providing proper protection.	3.12.4
29. Electrical equipment not prepared to receive 440/3/60 service.	3.15.1
30. Service connection at unit cooler not where shown on referenced drawing.	3.15.1
31. Electrical connections not clearly marked.	3.15.2
32. Electrical enclosures not splash proof or in accordance with MIL-E-24142.	3.15.3, 3.15.3.1, 3.15.3.2
33. Indicating lights missing or not as specified.	3.15.4
34. MANUAL-OFF-AUTO mode switch not operating as specified or missing.	3.15.6, 3.15.6.1, 3.15.6.2, 3.15.6.3
35. Cooling control missing or not as specified.	3.15.7, 3.15.7.1, 3.15.7.2, 3.15.7.3
36. Automatic electric defrost control system missing or not as specified.	3.15.8
37. Heater elements not in direct contact with the finned surface.	3.15.10 and 3.5
38. Heater elements not serviceable in the maintenance envelope shown on the referenced drawing.	3.11
39. Markings and identification plate missing, incomplete, not permanent, illegible, or not as specified.	3.16
40. Workmanship not conforming to the specification.	3.18

4.6.2 Capacity test. The capacity of the unit cooler shall be determined at the conditions specified in 3.4. The capacity test for the unit cooler shall be conducted in accordance with ASHRAE 25.

4.6.3 High-impact shock test. The shock test shall be as specified in MIL-S-901 for type A equipment. Evidence of fragmentation or missile effects of parts, deformation of parts that will cause active interference between parts, or coil leakage shall be cause for rejection. The unit cooler shall be operated continuously for a period of 72 hours after this test and before disassembly. During the 72 hours of operation, the capacity of the unit cooler shall not drop below 95 percent of rating.

4.6.4 Environmental vibration test. The environmental vibration test shall be conducted as specified in MIL-STD-167-1 for type I equipment. The operation of the unit cooler shall be unaffected by this test.

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4.6.5 Combination of list, trim and roll. The unit cooler shall operate for the conditions of list, trim, and roll specified in 3.8.

4.6.6 Leakage tests. The coil shall be charged with sufficient refrigerant to create a pressure of 20 lb/in^2 gauge; the coil shall then be tested for leaks with an automatic halogen leak detector. After this test a sufficient amount of dry nitrogen shall be introduced to create a pressure of 375 lb/in^2 gauge, and again the coil shall be tested with an automatic halogen leak detector. The automatic halogen leak detector used in these tests shall have a sensitivity for detecting flow of 1/2 ounce per year. After leak testing, the coil pressure shall be raised to 400 lb/in^2 gauge and held for 6 hours. No loss in pressure shall be permitted except as corrected for temperature. When the leakage and pressure tests have been completed, the coil shall be evacuated by means of a vacuum pump to 0.2 inch of mercury absolute and held for 1 hour. No change in vacuum shall be permitted except as corrected for temperature.

4.6.7 Fan test. The volume of air delivered through each unit cooler with louvers and fan guard installed shall be determined in accordance with AMCA Bulletin 210.

4.6.8 Motor test. The insulation resistance test and the dielectric strength test shall be conducted as specified in MIL-M-17059 and as applicable.

4.6.9 Fan blade balance test. The fan blade manufacturer shall certify through tests that the fan blades meet the requirements of 3.12.2.

4.6.10 Splash test. Electrical enclosures shall be splash-tested to ensure that the requirements of 3.15.3 are met. There shall be no signs of water entry into the enclosure.

4.6.11 Electrical shock hazard tests. The following tests shall be performed to prevent electrical shock.

4.6.11.1 Leakage current tests. A leakage current test shall be conducted on the unit cooler in accordance with MIL-E-917. Leakage current shall not exceed 5 milliamperes.

4.6.11.2 Insulation resistance test, unit cooler. The insulation resistance test shall be conducted on the unit cooler in accordance with MIL-P-24423. The insulation resistance shall be not less than that specified by MIL-P-24423.

4.6.11.3 Insulation resistance test, heating elements. The insulation resistance test for the heating elements shall be conducted by cycling the heater for 30 days submerged in water. In the test, the water shall be maintained at a temperature not less than that measured on the heater terminal seal or case material during heater operation, but not greater than 194°F. The heater shall be cycled four times per hour with an "ON" time of approximately 1-1/2 minutes and an "OFF" time of approximately 13-1/2 minutes. The insulation resistance measurement shall be made using a direct potential of 500 volts applied for a 1-minute period.

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The insulation resistance shall be not less than 50,000 ohms. At the conclusion of this test the heating element shall not break down when a dielectric strength test in accordance with 4.6.11.4 is applied.

4.6.11.4 Dielectric strength test. The contractor shall conduct a dielectric test in accordance with MIL-P-24423 on each unit cooler offered for delivery.

4.6.12 Internally excited vibration test. Fan assemblies shall be subjected to the vibration test in accordance with MIL-STD-167-1 to determine conformance to the requirements specified in 3.12.

4.6.13 Airborne noise test. Unit coolers and controls shall be subjected to the airborne noise test in accordance with MIL-STD-740-1 to determine conformance to the requirements specified in 3.13.

4.6.14 Operational test. Operational tests shall be performed to determine that the fan, heaters, and defrost controls function in accordance with the requirements of this specification. Tests shall be as follows.

4.6.14.1 Fan rated voltage test. The fan assembly of each unit cooler shall be tested at rated voltage and current to determine whether the fan will operate. Any unit cooler in which the fan or fan assembly fails to operate shall not be offered for delivery. Further, any unit cooler found to have excessive vibration during the start-up and operation of the fan and motor assembly shall not be offered for delivery.

4.6.14.2 Defrost test. The defrost system shall be initiated once by time clock and once manually, and in each case shall be cycled through one complete defrost period. The controls shall perform in accordance with the requirements of 3.15.8 through 3.15.8.3 and 3.15.9.

4.6.14.3 Sequence test. Four unit coolers shall be wired together as a system and cycled through defrost under the control of a master sequencer. The unit coolers shall perform as required by 3.15.8.1.

4.6.14.4 Heater element test. The heater elements shall be tested to ensure that the frost build-up on the coils can be melted within the requirements of 3.15.10 at minus 10°F.

4.6.14.5 Drainage test. During the operational test, the unit cooler shall be tilted 15 degrees in every direction to test that drainage conforms to the requirements of 3.9.1. The unit cooler shall then be tilted 30 degrees in all directions about a vertical centerline to test for spillage and dripping from the unit cooler casing. There shall be no spillage or dripping when the unit cooler is tilted in both tests.

4.6.14.6 Production reliability acceptance test. A Production Reliability Acceptance Test (PRAT) shall be conducted in accordance with MIL-STD-785, task 304, to ensure that the unit cooler meets the requirements of 3.2 (see 6.3). The unit cooler shall be operated for 500 hours with a 24-hour non-operative period

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every 100 hours of operation. (For the quality conformance inspection the unit cooler shall be required to operate only for one 100-hour period.) The unit cooler shall be operated in all conditions of list, trim, roll, and pitch in accordance with 3.8. The unit cooler shall be in the AUTO mode and set to automatically defrost for every 90 minutes of run time during the test. The unit cooler shall be operated in the MANUAL mode a minimum of 5 cycles. The unit cooler shall be cycled to the OFF mode for quality conformance test. This test may be run concurrently with any other applicable test.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of PPP-F-320.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2) and as follows:

(a) Cleaning.

(1) Clean refrigerant circuit internally after leak testing. Fill with trichlorotrifluoroethane solvent, conforming to MIL-C-81302, type II. Circulate solvent through the refrigerant piping system by means of an external pump. The

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pump shall have adequate capacity to circulate the solvent at a minimum velocity of 20 feet per second (ft/sec) (6.1 meters per second (m/sec)) through 3/8 inch (9.5 millimeter (mm)) outside diameter (od) liquid lines.

(2) Remove refrigerant control valves prior to filling the system with solvent. Open manual operating devices, if provided, during the cleaning period.

(3) Provide a circulating pump, strainers, filter and expansion tank. Install strainer with a muslin bag ahead of pump suction. Install a refrigerant filter with replaceable filter cores or cartridges between the pump discharge and the connection to the refrigerant system. The filter cores or cartridges shall be rated at 15 micrometers absolute (remove particles larger than 15 micrometers).

(4) Stop circulation of the cleaning solvent every 2 hours, and check the muslin bag in the strainer and the filter cores or cartridges in the refrigerant filter for accumulation of contaminants and foreign particles. Replace bags, cores or cartridges as necessary. Consider system to be clean when there is no evidence of contamination or foreign particles on the filter cores or cartridges after the 2-hour cleaning period.

(5) After cleaning units, remove the solvent, then blow out with dry nitrogen.

(6) The expansion valve shall be returned to normal and the unit evacuated and then pressurized with the holding charge of R-12 to 2 lb/in^2 and sealed for shipping.

5.2.1 Level A. Coolers shall be preserved in accordance with MIL-STD-2073-1; appendix C, table I. Each cooler shall be cushioned and placed in unit containers.

5.2.1.1 Unit pack. Unit packs (unit containers) shall conform to MIL-STD-2073-1; appendix F, table I. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option. Containers shall be of type weather resistant and conform to closure method V in accordance with the appendix of the box specification.

5.2.2 Level C. Coolers shall be preserved as specified for level A except that the unit containers specified in 5.2.1.1 shall be of the non-weather resistant type, class, or variety as applicable. Fiberboard box closure shall be in accordance with method I of the box specification, using pressure sensitive tape.

5.2.3 Commercial. Commercial preservation shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

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5.3.1 General. Shipping containers shall contain identical quantities of identical material and shall be of minimum weight and cube, similar construction, and of uniform size.

5.3.2 Level A, B, and C containers. Material preserved as specified (see 5.2) shall be packed in shipping containers for the level of packing specified (see 6.2) in accordance with MIL-STD-2073-1, appendix C, table VII. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Closure, gross weight, and waterproofing.

5.3.2.1.1 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that class weather resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding, and non-weather resistant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.2 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.2.1.3 Waterproofing. Unless otherwise specified (see 6.2), level A and, when specified (see 6.2), level B shipping containers shall be provided with caseliners, linings, wraps, or shrouds in accordance with the waterproofing requirements of MIL-STD-1186.

5.3.3 Commercial. Material preserved as specified (see 6.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with the minimum of 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base that will support the material and facilitate handling by mechanical handling equipment during shipment.

5.4 Marking. In addition to any special marking required (see 6.2), level A, B, and C interior packs and shipping containers shall be marked in accordance with MIL-STD-2073-1, appendix F, and commercial interior packs and shipping containers shall be marked in accordance with ASTM D 3951. In addition, bar coding shall be applied in accordance with the marking requirements of MIL-STD-2073-1.

5.5 Technical manuals. Technical manuals which accompany shipments shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed with any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall state the approximate location therein. The manual shall be readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specifications.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification describes unit coolers that are intended for use in ship's refrigerated stores compartments.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Sizes of unit coolers to be supplied (see 1.2 and 3.4).
(d) If first article inspection is required (see 3.1).
(e) Materials if other than those found in table I (see 3.3).
(f) Cooling coil connection hand (see 3.10).
(g) Airborne noise test (see 3.13).
(h) Structureborne noise criteria, if required (see 3.14).
(i) Whether the refrigerant solenoid valve will be supplied to the contractor (see 3.15.7.2).
(j) Quantity of heating cable (see 3.15.10).
(k) If inspection conditions are other than specified (see 4.3).
(l) Whether to accomplish production reliability acceptance test (see 4.6.14.6) for quality conformance inspection (see table II).
(m) Size of forced air cooler assemblies to be first article inspected (see 4.4).
(n) Structureborne noise test method (see table II).
(o) When lumber and plywood do not require fire-retardant treatment (see 5.1.1).
(p) Level of preservation required (see 5.2 and 5.3.3).
(q) If container selection is not at the contractor's option (see 5.2.1.1 and 5.3.2).
(r) Level of packing required (see 5.3 and 5.3.2).
(s) Whether shipping containers should be provided with waterproofing (see 5.3.2.1.3).
(t) Special marking required (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.5	DI-SAFT-80101	System safety hazard analysis report	-----
3.5	DI-E-7031	Drawings, engineering and associated lists	-----
4.3.1	DI-T-23731	Notification of tests	-----
4.4	DI-T-4902	First article inspection report	-----
4.6.14.6	DI-RELI-80250	Reliability test plan	-----
4.6.14.6	DI-RELI-80251	Reliability test procedures	-----
4.6.14.6	DI-RELI-80252	Reliability test reports	-----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ____ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Subject term (key word) listing.

Air unit cooler
Cooler
Refrigeration
Unit cooler

Preparing activity:
Navy - SH
(Project 4110-N417)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-24746(SH)	2. DOCUMENT DATE (YYMMDD) 31 DEC 1990

3. DOCUMENT TITLE
COOLERS, UNIT FORCED AIR FOR SHIP'S REFRIGERATED STORES

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contract (TPOC): Mr. J. Merold (SEA 56Y15) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 703-602-2073
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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